

DS 2500 *Fall* 2025 Lab Exercise 10

Deadline : 11/17/2025 11:59 PM EST

Late Deadline: 11/24/2025 11:59 PM EST with 50 points off

Total Points: 200 pts

Grading:

- This lab assignment is auto-graded. Gradescope can be a little picky, so make sure you don't put extraneous characters or whitespace in your answers -- and double-check the correct/incorrect confirmation.

Submission Instructions:

- Submit the Python file (.py) containing your solutions to Gradescope by the deadline specified at the top.
- You're allowed *unlimited* submissions.
- There are two deadlines. Submissions will not be accepted after the late deadline.

Disclosure: We've adapted lab problems from DS 2500 Spring 2025, so one or more problems may be the same or similar to those from that course.

Setup Instructions

- Use the following code to begin this lab

```
import pandas as pd
from sklearn.metrics import classification_report, mean_squared_error,
r2_score
from sklearn.neighbors import KNeighborsClassifier,
KNeighborsRegressor
from sklearn.model_selection import train_test_split

data = {
    'hours_worked': [155, 68, 172, 49, 28, 5, 160, 51, 163, 130, 118,
111, 47, 149, 116, 52, 175, 106, 65],
    'coffee_intake': [3.2, 2.8, 5.1, 2.4, 4.5, 1.9, 3.7, 2.9, 5.2,
2.3, 6.0, 3.1, 1.8, 5.8, 2.6, 4.7, 4.9, 5.3, 4.0],
    'stress_level': [6.1, 5.4, 7.8, 5.2, 6.9, 4.8, 6.0, 5.7, 7.4, 6.3,
8.2, 5.5, 4.9, 7.6, 5.8, 6.7, 7.3, 7.9, 6.1],
    'promotion': [1, 1, 0, 1, 1, 0, 1, 1, 0, 1, 0, 1, 1, 0, 1, 0, 0,
0, 0]
}
df = pd.DataFrame(data)
```

Problem 1 (50 pts)

- Use `train_test_split()` to split the dataframe `df` into 2 dataframes.
 - `test_size` must be set to `0.2`.
 - `random_state` must be set to `0`.
- The resulting dataframes should be named as `df_tr` and `df_te`.

Problem 2 (75 pts)

- Instantiate a KNN classifier as follows:

```
problem2_model = KNeighborsClassifier(n_neighbors=3)
```

- Train the model using the training data.
 - Features: `hours_worked`, `coffee_intake`
 - Target: `promotion`
- Make predictions on the test set.
- Generate and print a classification report using `classification_report()`.

Problem 3 (75 pts)

- Instantiate a KNN regressor as follows

```
problem3_model = KNeighborsRegressor(n_neighbors=3)
```

- Train the model using the training data.
 - Features: `hours_worked`, `coffee_intake`
 - Target: `stress_level`
- Make predictions on the test set.
- Evaluate the model performance using `mean_squared_error` and `r2_score`

Extra Problem 1 (0 pt; not graded)

1. True or False: A confusion matrix is used to evaluate classification models.
2. True or False: Linear regression is a supervised learning model.
3. True or False: K-Nearest Neighbors is an unsupervised learning algorithm.
4. True or False: In supervised learning, the dataset must contain target labels.
5. True or False: An R^2 score of 1 means that the model perfectly predicts the target variable.

Extra Problem 2 (0 pt; not graded)

1. Create a 2x2 confusion matrix with random numbers.
2. How many total samples are there? How many positive samples? what about negative?
3. Explain what each cell in the matrix represents.
4. Compute Accuracy, Precision, Recall based on the matrix.

Extra Problem 3 (0 pt; not graded)

1. List examples of targets we can predict using a linear regression model
2. List examples of targets we can predict using a logistic regression model
3. Describe the KNN algorithm
4. Given two data points: (3,0) and (0,4), compute Euclidean distance and Manhattan distance.
5. What is the difference between Euclidean and Manhattan distance?